The Quadratic Formula

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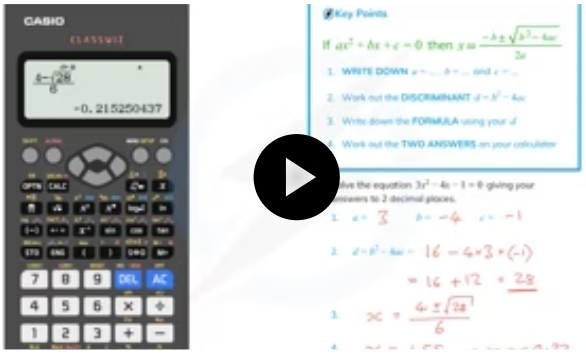

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Quadratic Formula

What is the quadratic formula?

- A **quadratic equation** has the form $ax^2 + bx + c = 0$ (where $a \neq 0$)
  - you need "**= 0**" on one side
- The **quadratic formula** is a formula that gives both solutions to a quadratic equation:

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Examiner Tips and Tricks

- Make sure the quadratic equation has "= 0" on the right-hand side
  - Otherwise it needs rearranging first

Stuck on something?
Ask questions and get help with exactly what you're revising.
Got it
Help me understand

How do I use the quadratic formula to solve a quadratic equation?

- Put **brackets** around any **negative numbers** being substituted in

- To solve $2x^2 - 8x - 3 = 0$ using the quadratic formula:
  - $a = 2$, $b = -8$ and $c = -3$
  - $x = \frac{-(-8) \pm \sqrt{(-8)^2 - 4 \times 2 \times (-3)}}{2 \times 2}$
  - Type this into a calculator or simplify by hand
    - Type it once using + for ± then again using - for ±
  - The **solutions** are $x = 4.3452078...$ or $x = -0.34520787....$
    - To 3 **decimal places**: $x = 4.345$ or $x = -0.345$
    - To 3 **significant figures**: x = 4.35 or x = -0.345

Examiner Tips and Tricks

- Always look for how the question wants you to leave your final answers
  - For example, correct to 2 decimal places

How do I write the solutions in an exact (surd) form?

- You may be asked to give answers in an **exact** (**surd**) form
  - For example, in a non-calculator paper

- In the example above, work out the **number** under the **square root** sign
  - Be careful with negatives!
    - $(-8)^2 - 4 \times 2 \times (-3) = 64 + 24 = 88$
  - Now square root this number and use **surd rules** to simplify
    - $\sqrt{88} = \sqrt{4 \times 22} = \sqrt{4} \times \sqrt{22} = 2\sqrt{22}$
  - **Substitute** this back into the formula and simplify
    - $x = \frac{8 \pm 2\sqrt{22}}{4} = \frac{2\left(4 \pm \sqrt{22}\right)}{4} = \frac{4 \pm \sqrt{22}}{2}$
    - The solutions in **exact** (**surd**) form are $x = \frac{4 + \sqrt{22}}{2}$ or $x = \frac{4 - \sqrt{22}}{2}$

- **Calculators** that can solve quadratics will give solutions in **exact** (**surd**) form

What is the discriminant?

- The part of the formula under the square root ($b^2 - 4ac$) is called the **discriminant**
- The **sign** of this value tells you if there are 0, 1 or 2 solutions
  - If $\boldsymbol{b^2 - 4ac > 0}$ (positive)
    - then there are 2 different solutions
  - If $\boldsymbol{b^2 - 4ac = 0}$

- then there are no solutions
- If your calculator gives you solutions with $\mathbf{i}$ terms in, these are "complex" and are not what we are looking for

- Interestingly, if $b^2 - 4ac$ is a perfect square number (1, 4, 9, 16, ...) then the quadratic expression could have been factorised!

Worked Example

Use the quadratic formula to find the solutions of the equation $3x^2 - 2x - 4 = 0$.
Give each solution as an exact value in its simplest form.

Write down the values of a, b and c

$$a = 3, b = -2, c = -4$$

Substitute these values into the quadratic formula,

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

Put brackets around any negative numbers

$$x = \frac{-(-2) \pm \sqrt{(-2)^2 - 4 \times 3 \times (-4)}}{2 \times 3}$$

Simplify the expressions

$$x = \frac{2 \pm \sqrt{4 + 48}}{6} = \frac{2 \pm \sqrt{52}}{6}$$

Simplify the surd

$$x = = \frac{2 \pm \sqrt{4 \times 13}}{6} = \frac{2 \pm 2\sqrt{13}}{6}$$

Simplify the fraction

$$x = \frac{1 \pm \sqrt{13}}{3}$$

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